NextGenC — Gridded SOC maps of Finland

Procedural note for the kriged SOC map products that accompany the NextGenC report. Companion to build_soc_matrices.R (point matrices) and build_soc_maps.R (this product).

1 What this produces

For each of the 6 models (SP1, TP2, TP3, Yasso07, Yasso15, Yasso20), the posterior-predictive SOC at the $\sim$447 calibration plots is interpolated to a regular grid covering Finland, **one layer per year, 1985–2024 (40 years)**. Two multiband GeoTIFFs per model, plus a multi-model **ensemble** pair (**14 files total**):

File	Bands	Content
<MODEL>_SOC_mean.tif	40	Kriged mean total SOC (tC/ha), band n = year 1984+n
<MODEL>_SOC_sd.tif	40	Total uncertainty SD (tC/ha), same grid/year as mean band
ENSEMBLE_SOC_mean.tif	40	Six-model ensemble mean SOC (tC/ha)
ENSEMBLE_SOC_sd.tif	40	Ensemble total SD (within + between-model + spatial)

Band names carry the year (e.g. y1985), written into GeoTIFF metadata so labels travel with the file. Band n of every SD stack is the uncertainty for band n of the matching mean stack (same year, grid, and land mask).

2 Grid and storage

- **CRS:** ETRS-TM35FIN (EPSG:3067), metres.
- **Resolution:** 2 km. Grid 360 $\times$ 600 ($\sim$216 k cells, of which $\sim$42 % land).
- **Mask:** clipped to the Finland boundary from maakunta_sf.rds.
- **Format:** GeoTIFF, DEFLATE-compressed, float32.
- **Footprint:** $\sim$0.12 GB for all 560 maps (14 files).

The 2 km choice is a deliberate compromise: the $\sim$447 plots have a **mean spacing of $\sim$27.5 km**, so 2 km already oversamples the information content — finer grids render smoother but add no real spatial detail. Disk budget (1–2 GB) is not the binding constraint; data density is.

3 Method

Kriging is done **in log space** and back-transformed, because SOC is positive and right-skewed and HIKET's error model is multiplicative — this keeps the map uncertainty multiplicative and consistent with the calibration. Per model $\times$ year:

1. **Inputs.** Plot mean and SD come from the posterior-predictive bundles (the same soc_mean / soc_sd columns used by build_soc_matrices.R); plot coordinates (x_ETRS, y_ETRS) come from Data/model_inputs/site_ joined on plot_id.
2. **Krige the mean.** Ordinary kriging of log(soc_mean) over the grid $\rightarrow$ interpolated mean surface plus the **kriging prediction variance** v_{krige} (small near plots, large in data-sparse areas, e.g. Lapland).
3. **Krige the model uncertainty.** Ordinary kriging of log(soc_sd) $\rightarrow$ a surface of the model's own posterior-predictive uncertainty.
4. **Combine (option C).** Total uncertainty fuses both sources,

$$\text{sd}_{\text{total}} = \sqrt{v_{\text{krige}} + \text{kriged\_modelSD}^2}$$

(combined in log space, then back-transformed). This propagates **model uncertainty** *and* **spatial-coverage uncertainty** into a single SD layer — neither source is silently dropped.

5. **Back-transform & mask.** Exponentiate to tC/ha, clip to the land mask, write the two stacks.

Variogram: **one pooled model per model**, reused for all 40 years (the spatial structure of SOC is essentially static year-to-year; this is stable and avoids per-year auto-fit failures). `set.seed(2025)` as elsewhere in HIKET.

Peat & water exclusion. The maps target **mineral-soil forest SOC**. Peat/water calibration plots (`site_raw` soil codes `Tu/Ve`) are dropped from the kriging *input* so they do not pull the interpolated surface, and grid cells are masked to NoData where they are >**50 % peatland** or >**50 % water**. Peat: Luke MS-NFI site-main-class raster (`paatyyppi`, 16 m, EPSG:3067; spruce/pine mire + treeless peatland), aggregated to the 2 km grid as a peat fraction *of the whole cell* (water in the denominator, so lake-dominated cells do not read as peat). This site-type definition captures drained peatland forest (*turvekangas*), which land-cover products miss (∼10 % of land masked). Water: SYKE Ranta10 lake polygons (the sea is outside the land polygon). See `Data/Peat_extension/` and `Data/Water_extension/`. In the thumbnail figures peat renders **black** and water/sea **white**.

4 Ensemble (six-model) product

The ensemble is formed *on the grid*, after each model has been kriged — i.e. the per-model mean and total-SD stacks above are combined cell-by-cell, equal-weighted ($w_i = 1/6$). With $M = 6$ model mean surfaces m_i and total-SD surfaces s_i (each s_i already folds in that model's kriging variance and posterior uncertainty, §3 step 4):

$$m_{\text{ens}} = \frac{1}{M}\sum_i m_i, \qquad s_{\text{ens}} = \sqrt{\underbrace{\tfrac{1}{M}\textstyle\sum_i s_i^2}_{\text{within-model}} + \underbrace{\tfrac{1}{M-1}\textstyle\sum_i (m_i - m_{\text{ens}})^2}_{\text{between-model (structural)}}}\,.$$

This is the law of total variance: the ensemble deviation fuses **(i)** each model's own predictive uncertainty (which already carries **(iii)** the spatial/kriging uncertainty) with **(ii)** the *structural spread* between models — the among-model disagreement that is the central HIKET question. The between-model term is the scientifically interesting one: where it dominates, the map is uncertain because the models *disagree*, not because the data are sparse. Combination is done in the same (log/back-transformed) space as the per-model SDs for consistency.

Note: the between-model term is also informative on its own; if a structural-disagreement map is wanted later, export $\sqrt{\tfrac{1}{M-1}\sum_i (m_i - m_{\text{ens}})^2}$ as a separate stack — one extra line in the script.

5 How to run

```
# from the project root
Rscript Reporting/NextgenC_report/build_soc_maps.R
```

Reads the same `bundles <- c(...)` RUN_ID map as `build_soc_matrices.R`. Writes the 12 GeoTIFFs into `Reporting/NextgenC_report/`.

Dependencies: `terra`, `gstat`, `sf` (and `sp`). Posterior-predictive bundles must be present in `Calibration_real_data_transient/runs/`, and `Data/model_inputs/{site_raw.csv, maakunta_sf.rds}` must exist.

6 Maintenance notes

- **TP3 RUN_ID is provisional.** As in `build_soc_matrices.R`, the TP3 bundle is the pre-fix (Euler) predictive. Swap the TP3 RUN_ID to the exact-integrator re-calibration when it is synced from Puhti, then re-run.

- **Adding years / plots.** No code change — the script reads year columns and plot rows from the bundles directly.
- **Changing resolution.** Edit the single `RES_M <- 2000` constant. Storage scales as $1/\text{res}^2$: 1 km ≈ 0.4 GB, 5 km ≈ 0.02 GB (all 560 maps, compressed).
- **Reading the output.** `terra::rast("SP1_SOC_mean.tif")` gives a 40-layer SpatRaster; layer names are the years. Same for the `_sd` file.